2024

MOLECULAR BIOLOGY

GROUP 7

INTRODUCTION

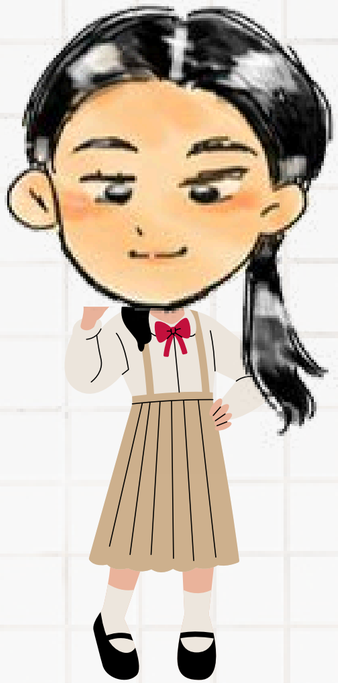
Bá Tuyên



Khánh Đan



Mai Anh

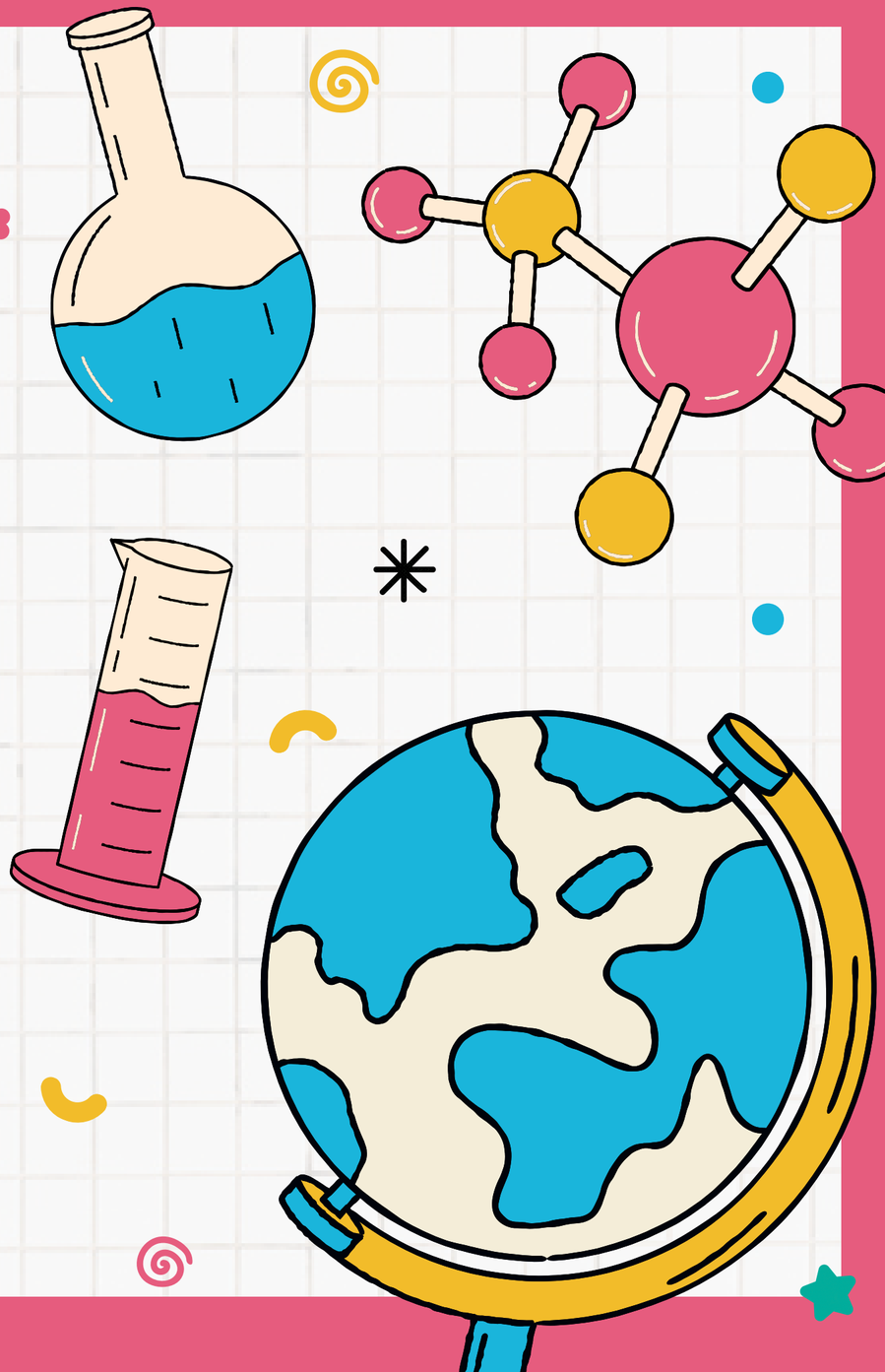


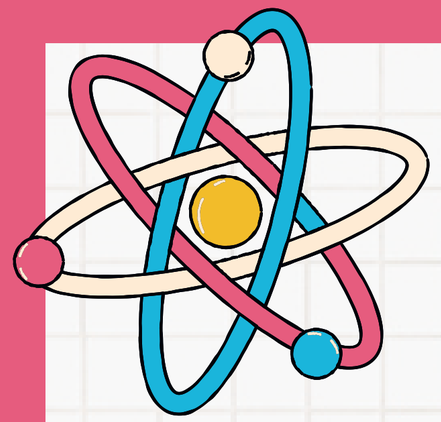
Anh Xuân



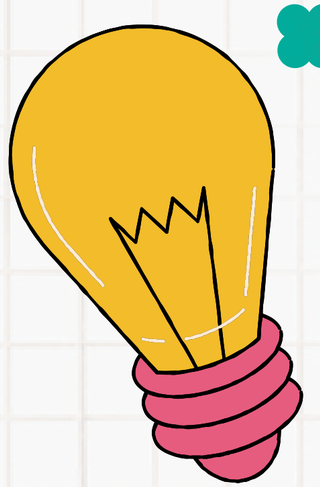
DAY 1

LAB SAFETY & PREPARATION





LAB SAFETY



I. UNDERSTANDING RULES

Understand and master laboratory regulations on experimental procedures and troubleshooting methods.

II. POTENTIAL RISKS

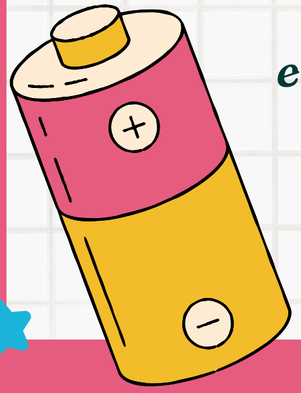
Keep the surrounding research environment clean and neat to avoid affecting the research process and achieve the best results.

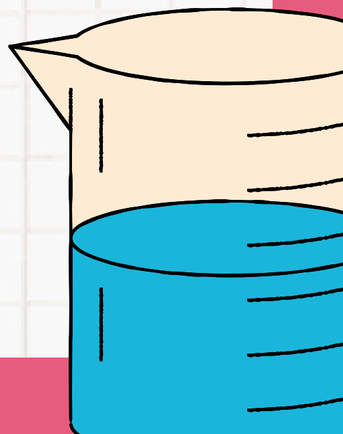
III. CLEAN AND NEAT

Know potential risks from materials, chemicals, and equipment to ensure personal safety

IV. DEVICE

Understand how equipment operates and use it correctly to achieve the best research results. Handle research equipment gently to avoid damage.





PREPARATION

TOOL PREPARATION

Prepare petri dishes, tubes for experiments, Erlenmeyer flasks, alcohol burners, glass slides, cover slips, distilled water, microbial samples, and inoculation loops



COTTON PLUG PREPARATION

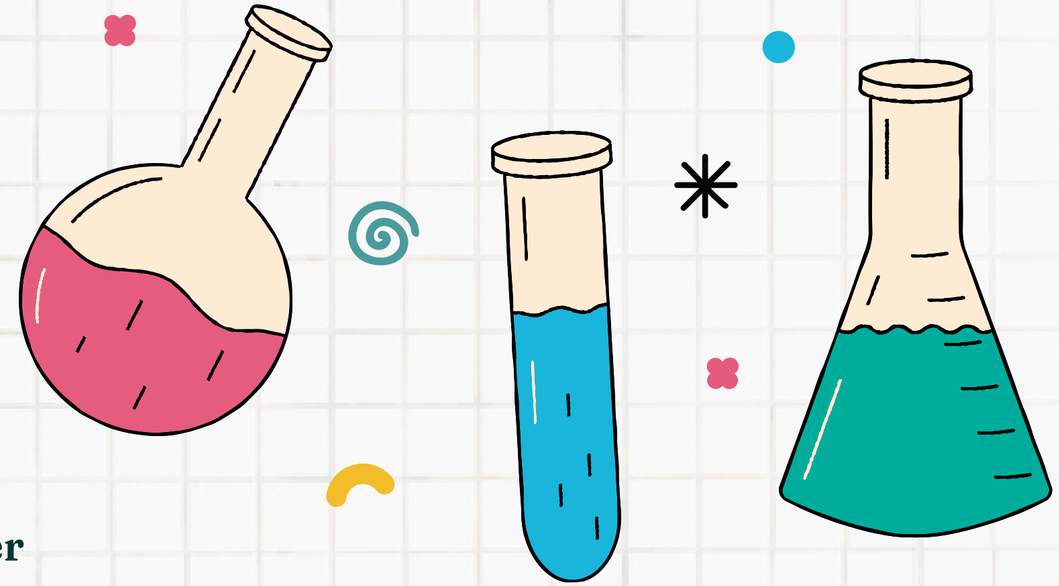
Put cotton in tubes and Erlenmeyer flasks to keep the internal solution isolated from the external environment.



STERILIZATION

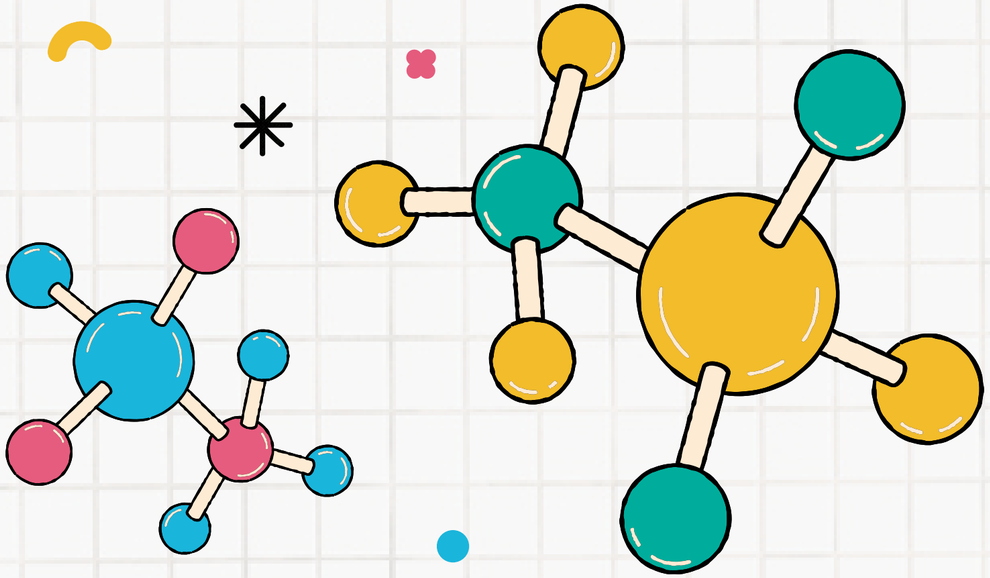
Place the cotton-plugged tubes and flasks into the Electrothermal Pressure Steam Sterilizer and sterilize to clean tools for the experiment.

01 LEAF SECTIONING



Thinly peel a leaf to obtain a thin section, prepare the sample:

- **Place on a glass slide**
- **Add distilled water**
- **Place a coverslip on top, then place under the microscope**

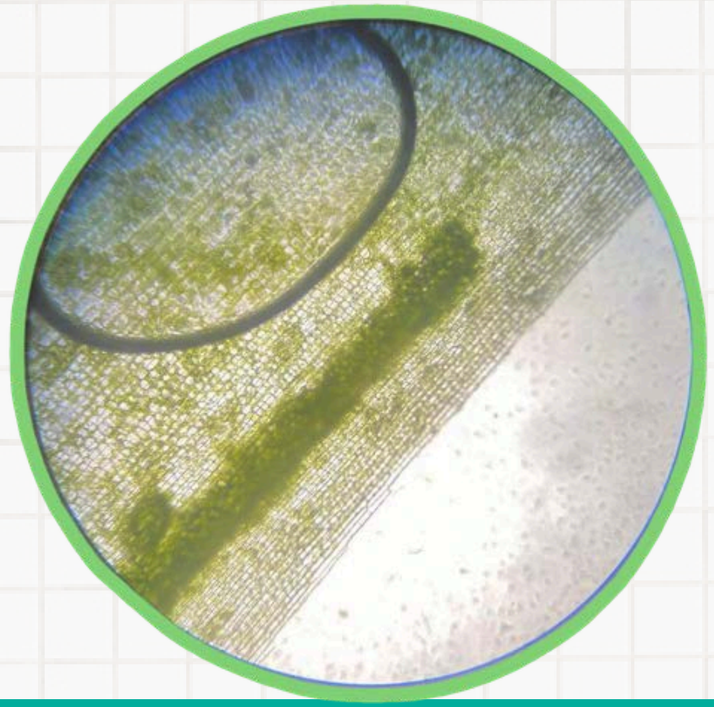


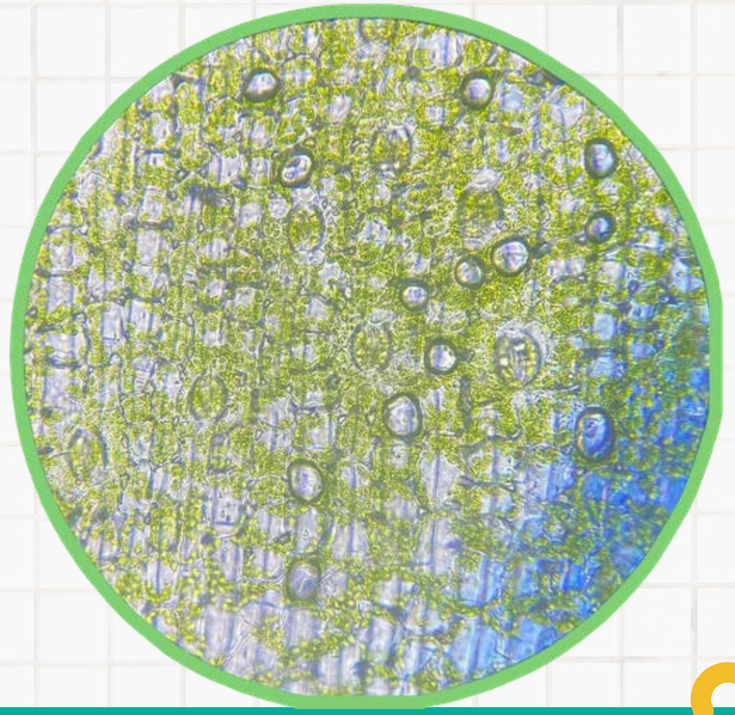
02 OBSERVATION

- **Turn on the microscope light**
- **Observe using objective lenses (adjust up/down, side-to-side for clarity)**
- **Switch objective lenses (rotate all lenses until clear)**
- **Observe and take photos of results**

RESULTS

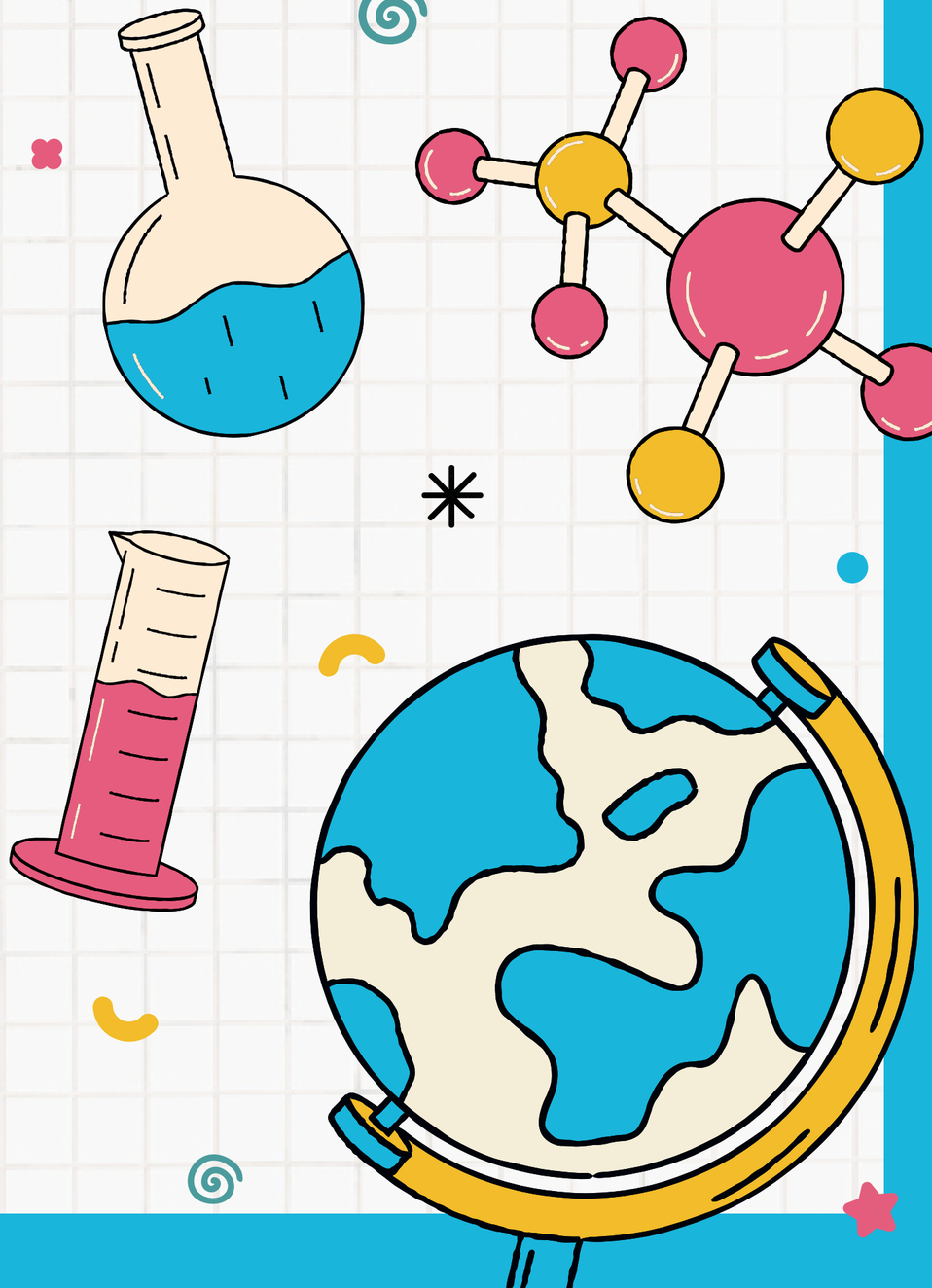
Chloroplasts visible on the leaf /
Stomata visible on the leaf





DAY 2

MICROBIAL ISOLATION



MEDIA PREPARATION

I. PREPARATION OF 10 PDA PETRI DISHES:

- Take 9.8g PDA into an Erlenmeyer flask
- Add 200ml distilled water (20ml per petri dish)
- Make a cotton plug for the flask
- Sterilize -> then divide into 10 dishes

 Purpose: Medium for culturing germs



II. PREPARATION OF 20 LB BROTH TEST TUBES

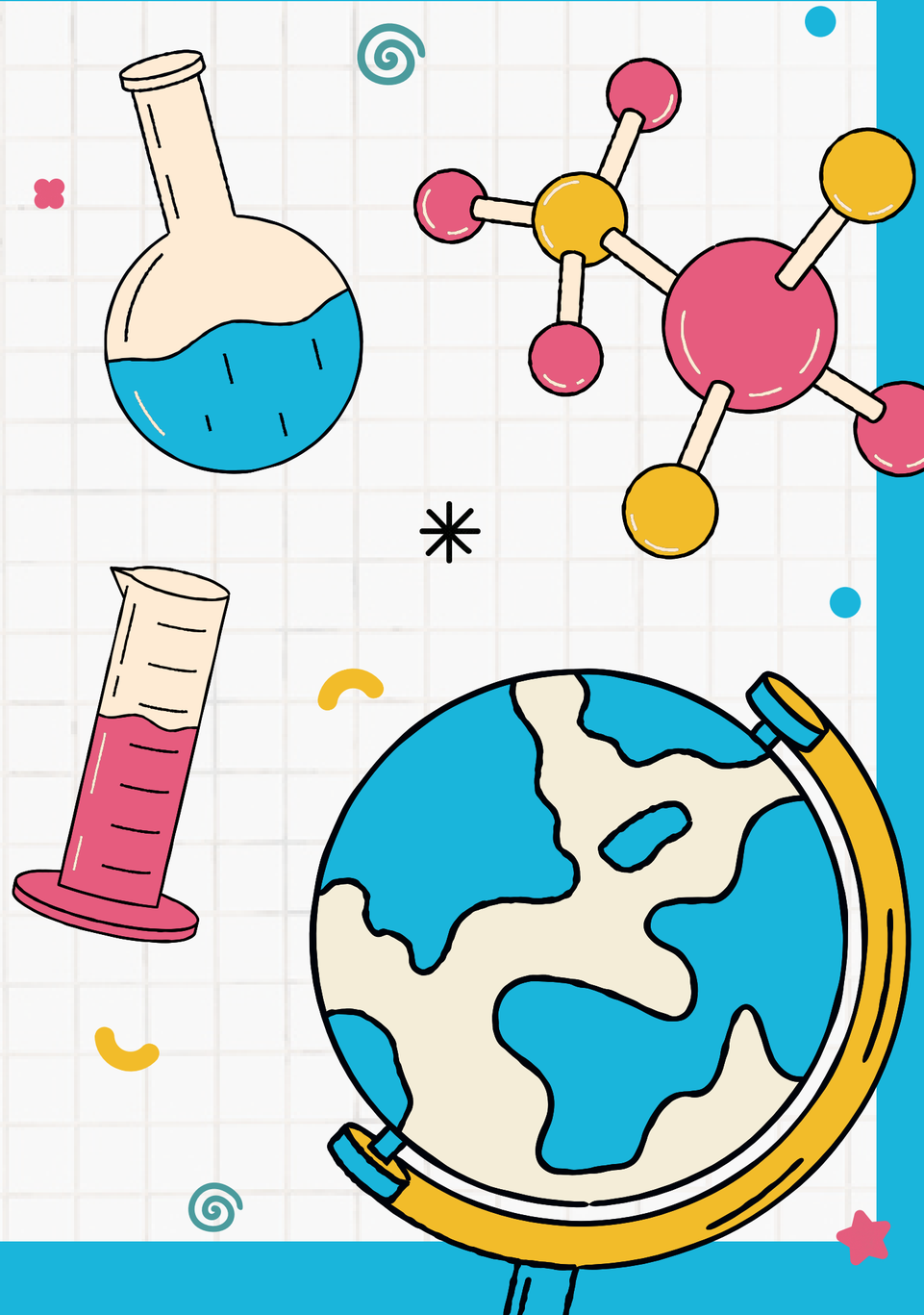
1. Take 2.25g LB Broth into tubes evenly distributed
2. Add 90ml distilled water (4.5ml per test tube)
3. Divide evenly into 20 test tubes
4. Make cotton plugs - wrap them up





DAY 3

SOIL SAMPLE INOCULATION



Samples: Phan Thiet 1, Phan Thiet 2

PROCEDURE:

01 TUBE 1 (10^{-1})

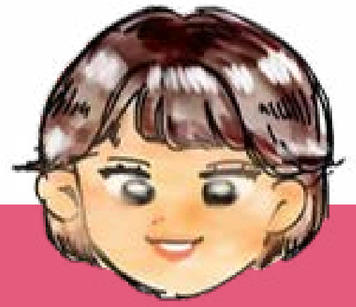
0.5g soil + 4.5ml distilled water

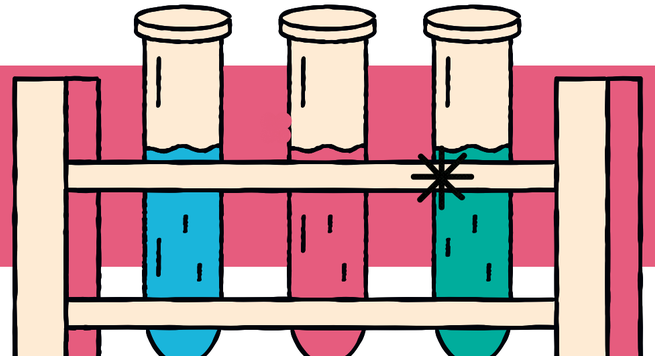
02 TUBE 2 (10^{-2})

0.5ml solution from tube (1)
+ 4.5ml distilled water

03 TUBE 3 (10^{-3})

0.5ml solution from tube (2)
+ 4.5ml distilled water







Sample Prep: PT1: 10^{-1} and 10^{-3} ; PT2: 10^{-1} and 10^{-3}

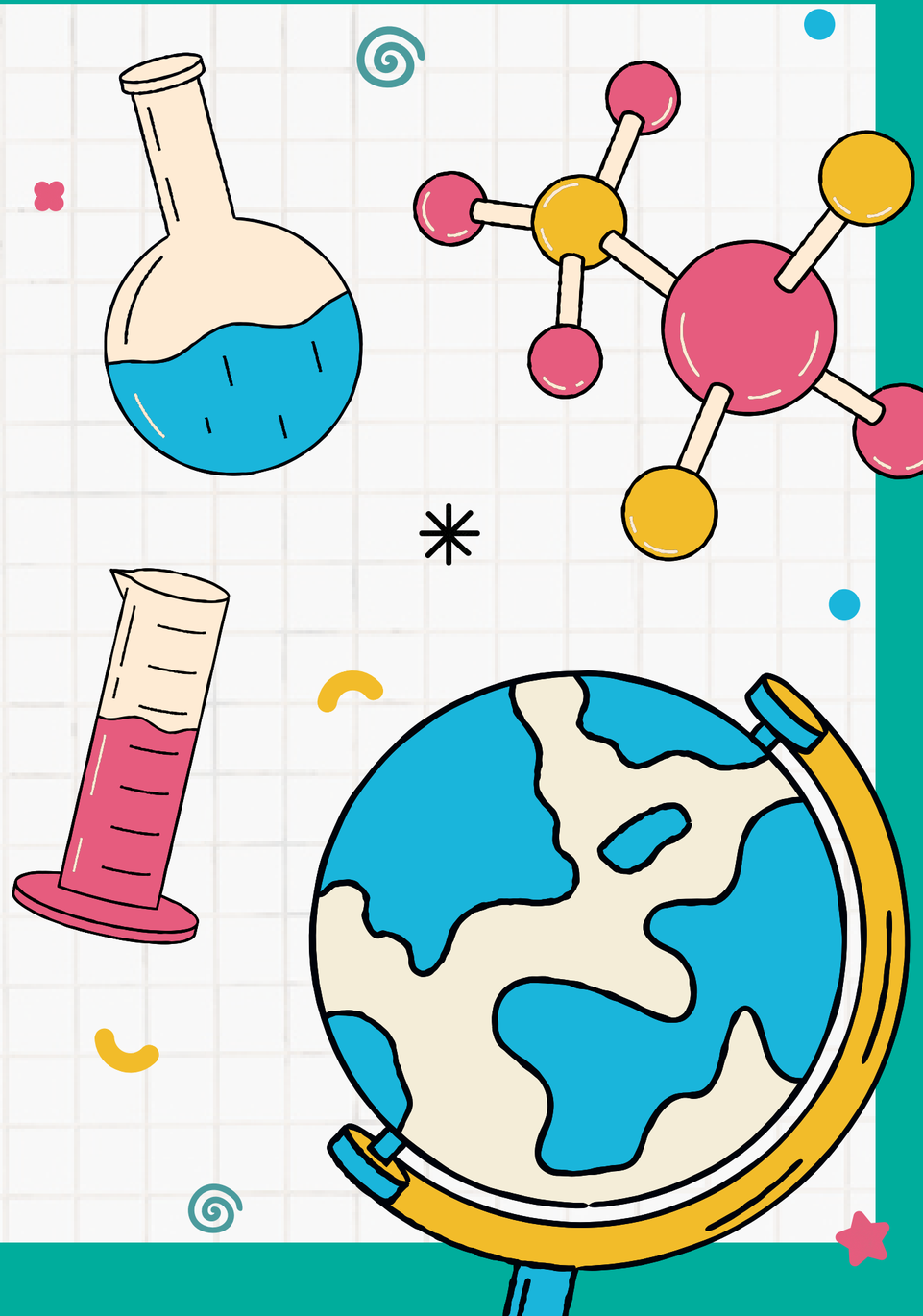
STEPS:

Step 1: Sterilization (10 minutes in boiling water bath)
Step 2: Inoculate germs from tube (1) and tube (3)

NOTES:

(1) Sample collection using a micropipette
(2) Use 98° alcohol to sterilize the inoculation loop, burn over flame
(3) Gently inoculate until the surrounding sample on the petri dish dries.

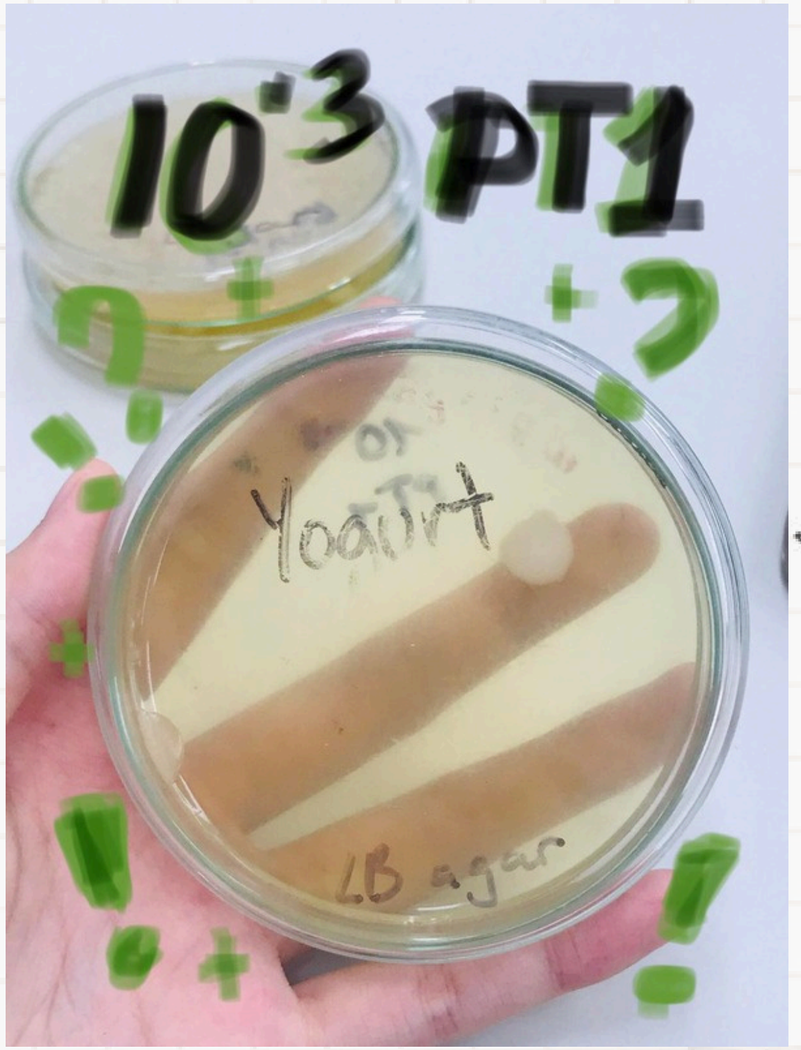
DAY 4-5





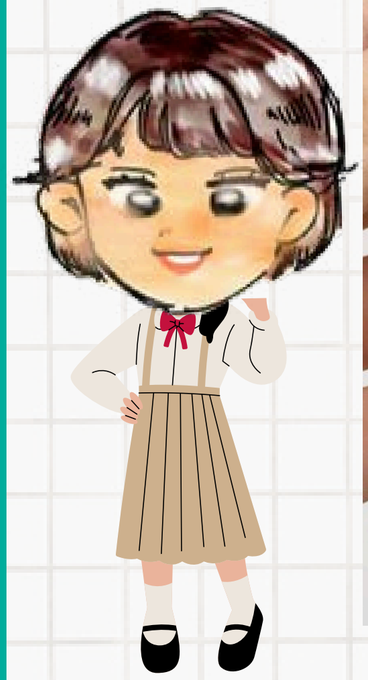


ANH XUÂN



BÁ TUYÊN



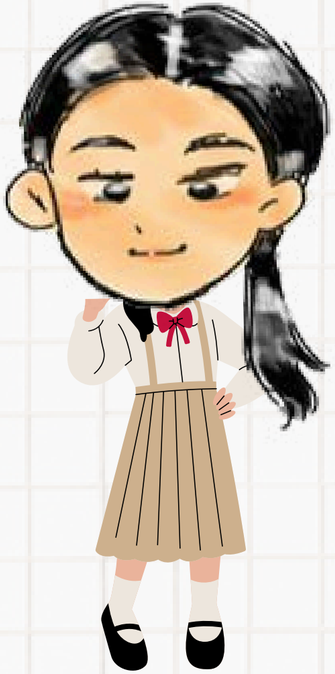




KHÁNH ĐAN



MAI ANH



I. MICROBIAL INOCULATION ON LB AGAR

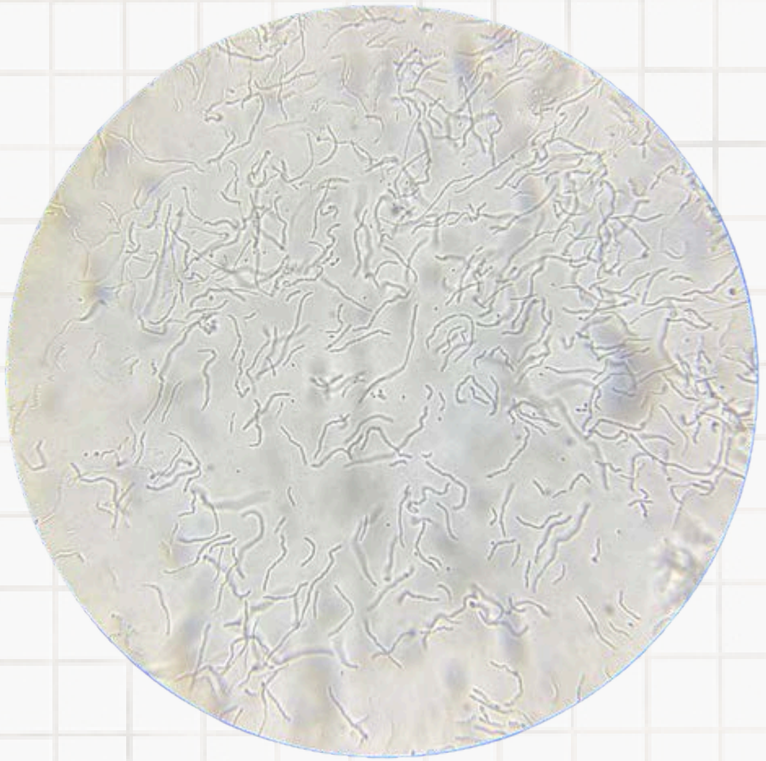
Use an inoculation loop to pick 5 different microorganisms, streaking 5 different lines on the LB Agar medium plate

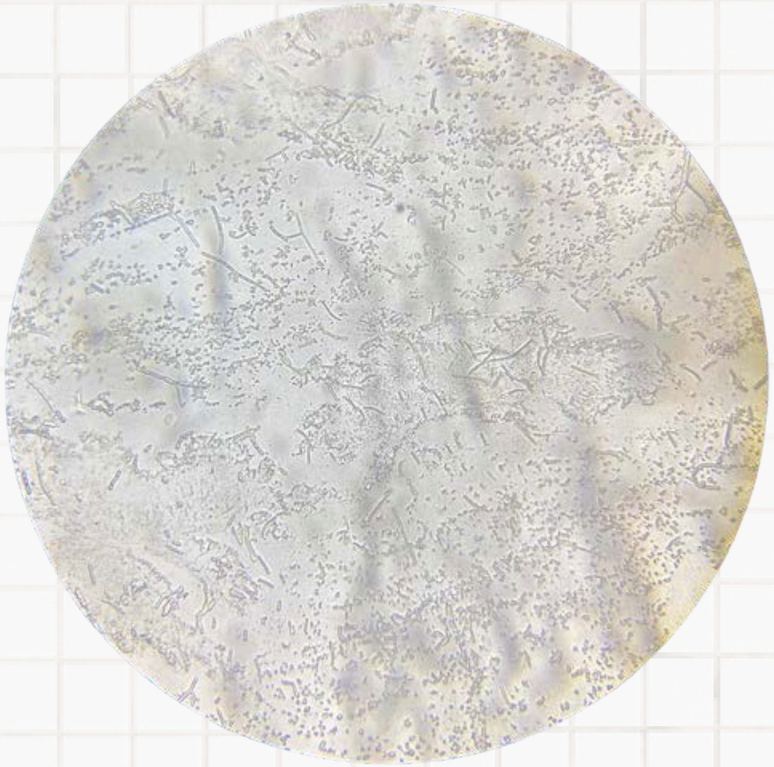




II. OBSERVATION

- Use an inoculation loop to take a sample from the dish in step 1 onto a coverslip
- Observe sequentially under 4 different objective lenses





III. DROPLET INOCULATION

1. Take 5 microorganisms and mix in 100µl distilled water

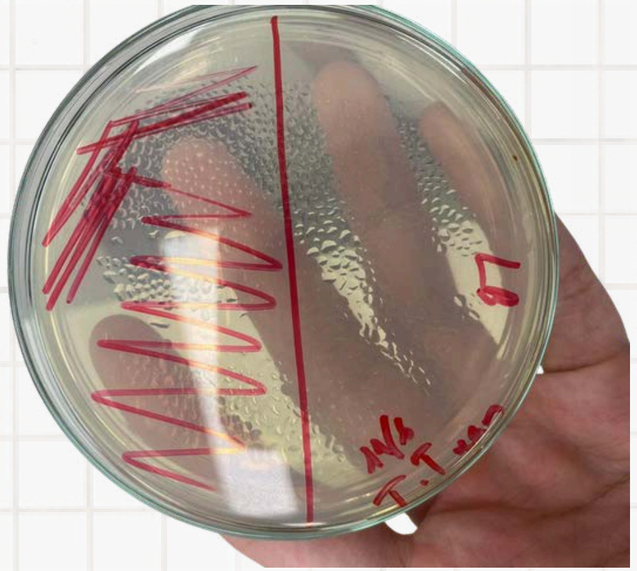
2. Prepare 3 petri dishes containing 3 different media: xylene, peptone, CMC

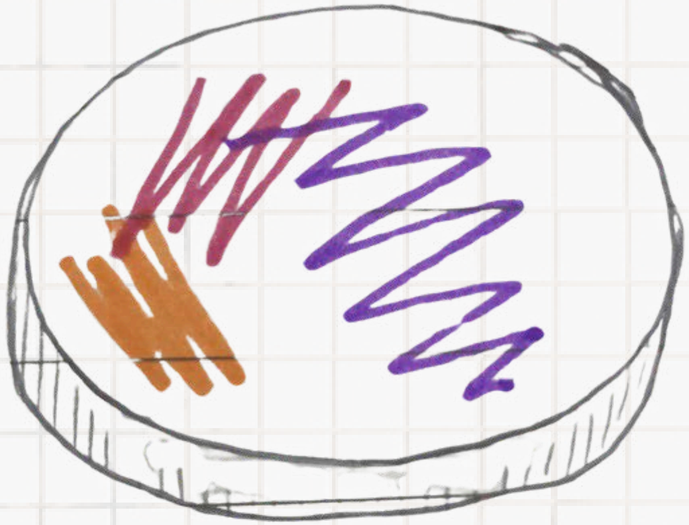
3. Drop 5 drops (10µl/drop) of the prepared sample above into 5 different spots on the medium. 5 drops per medium



IV. STREAK PLATE METHOD

- Divide the medium plate into 2 parts, streak each plate using the streak plate method (3-zone streaking)
- Using a sterilized inoculation loop, take samples from points 1, 3, 4, 5 (best-growing microorganisms)
- Streak the first line, then re-sterilize the inoculation loop, continue streaking the 2nd line starting from the 1st line, repeat similarly for the 3rd line
- Expected Result: The first zone has the highest microbial concentration, the 2nd zone is lighter, and the final zone has only isolated colonies

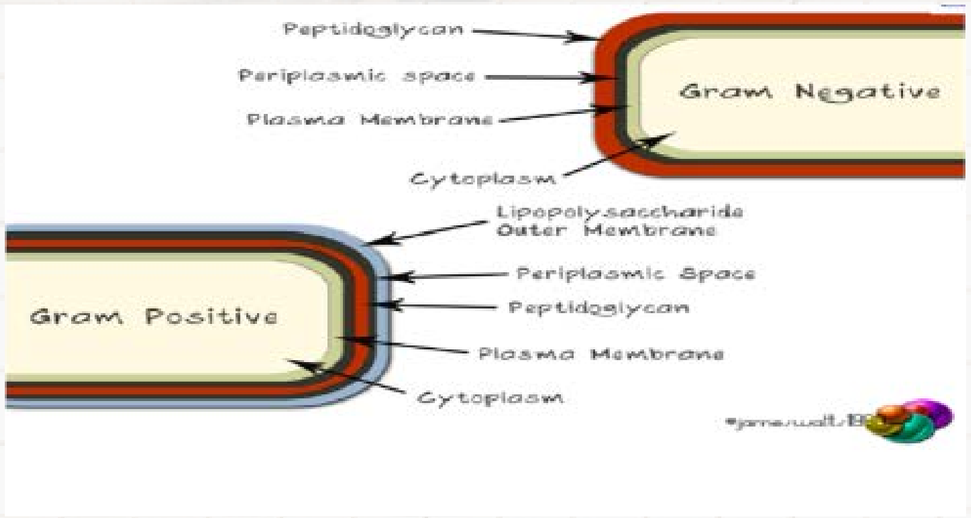




V. GRAM STAINING & STREAKING PRINCIPLES

- Bacteria are divided into 2 types: Gram-positive and Gram-negative
- Gram+: Thick cell wall
- Gram-: Thin cell wall

- Using double Gram staining helps differentiate them



VI. Double Gram Staining Procedure



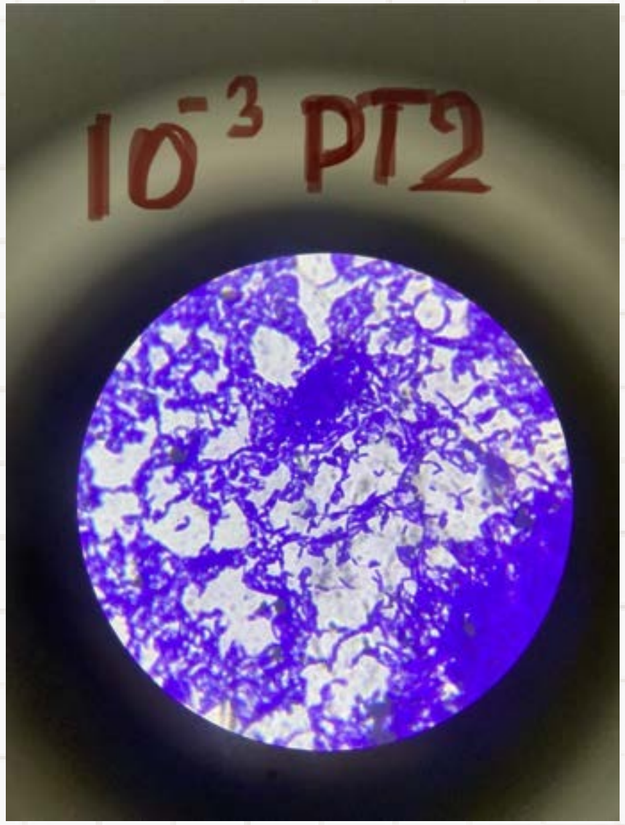
1. Place the microbial sample on a glass slide, use **crystal violet** for staining

2. Wait 1 minute for the stain to penetrate the sample, rinse with distilled water

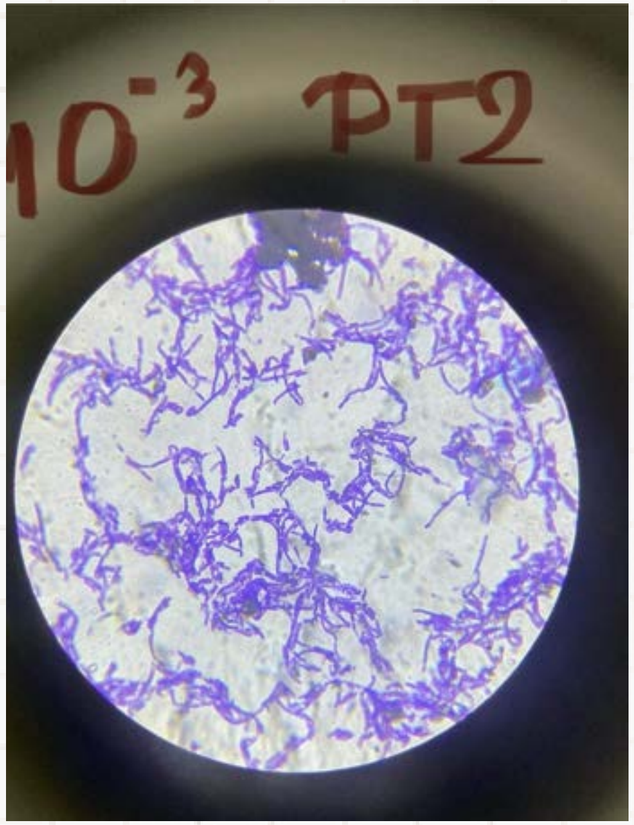
3. Mordant with **Lugol's iodine** wait 1 minute, then rinse with 95% alcohol.

4. Counterstain with **fuchsin**, wait 1 minute, rinse with tap water and observe under a microscope

Staining Results

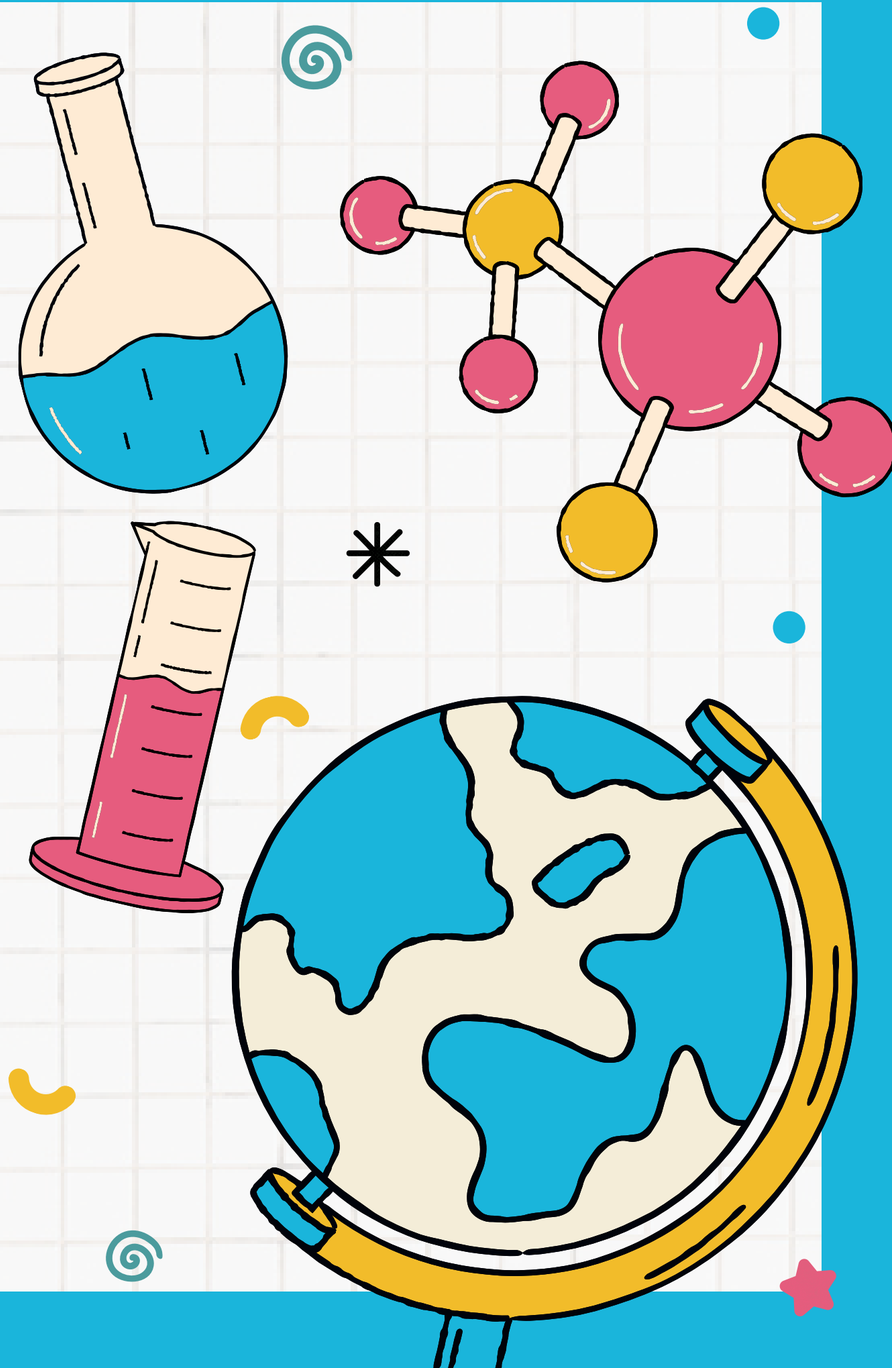






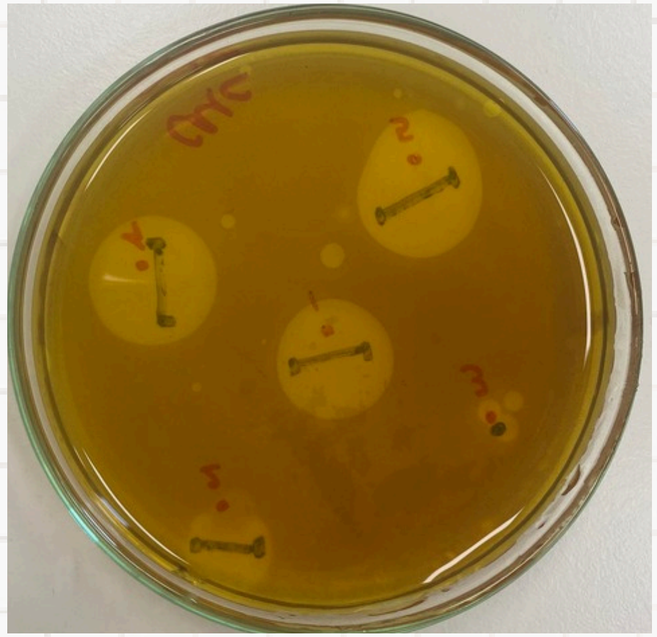
DAY 6-7

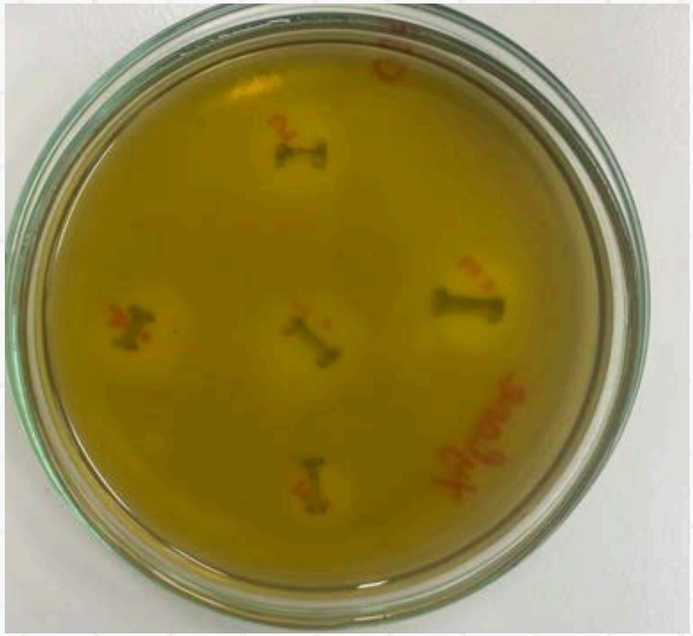
PCR - PRODUCT PURIFICATION & DIGESTION



ENZYME DIGESTION & CLEARING ZONE ANALYSIS

- Examine enzyme degradation and growth.
- Wash the surface with water
- Pour Lugol's iodine and observe enzyme clearing zones (diameters)
- Compare with microbial colony diameters





DIAMETER MEASUREMENT RESULTS

XYLANE

	D1 (mm)	D2 (mm)	D2 - D1 (mm)
Strains 1	10	11	11 - 10 = 1
Strains 2	8	10	10 - 8 = 2
Strains 3	11	14	14 - 11 = 3
Strains 4	6	14	14 - 6 = 8
Strains 5	8	13	13 - 8 = 5

LB AGAR

	D1 (mm)	D2 (mm)	D2 - D1 (mm)
Strains 1	22	27	27 - 22 = 5
Strains 2	18	28	28 - 18 = 10
Strains 3	15	17	17 - 15 = 2
Strains 4	11	17	17 - 11 = 6
Strains 5	14	25	25 - 14 = 11

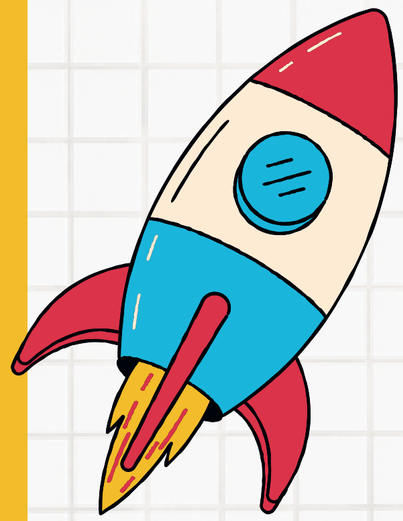
PECTIN

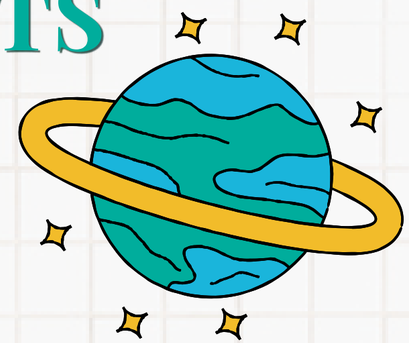
	D1 (mm)	D2 (mm)	D2 - D1 (mm)
Strains 1	6	8	8 - 6 = 2
Strains 2	0	0	0
Strains 3	0	0	0
Strains 4	3	4	4 - 3 = 1
Strains 4	6	7	7 - 6 = 1

CMC

	D1 (mm)	D2 (mm)	D2 - D1 (mm)
Strains 1	11	18	18 - 11 = 7
Strains 2	10	12	12 - 10 = 2
Strains 3	2	4	4 - 2 = 2
Strains 4	11	19	19 - 11 = 8
Strains 5	14	21	21 - 14 = 7

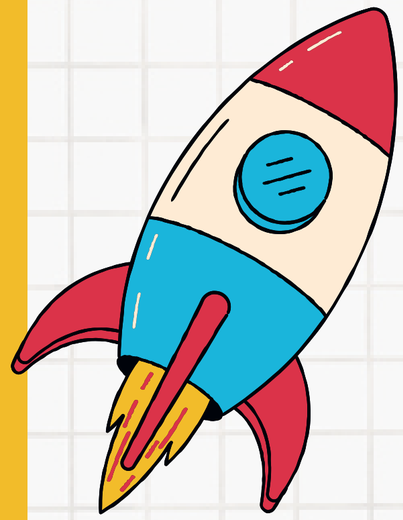
TURBIDITY MEASUREMENT RESULTS

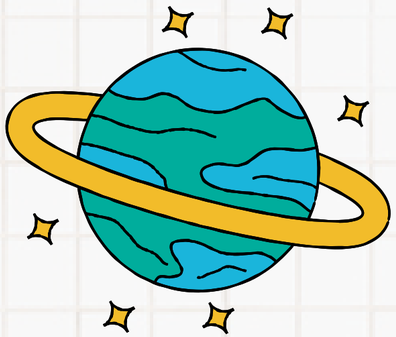


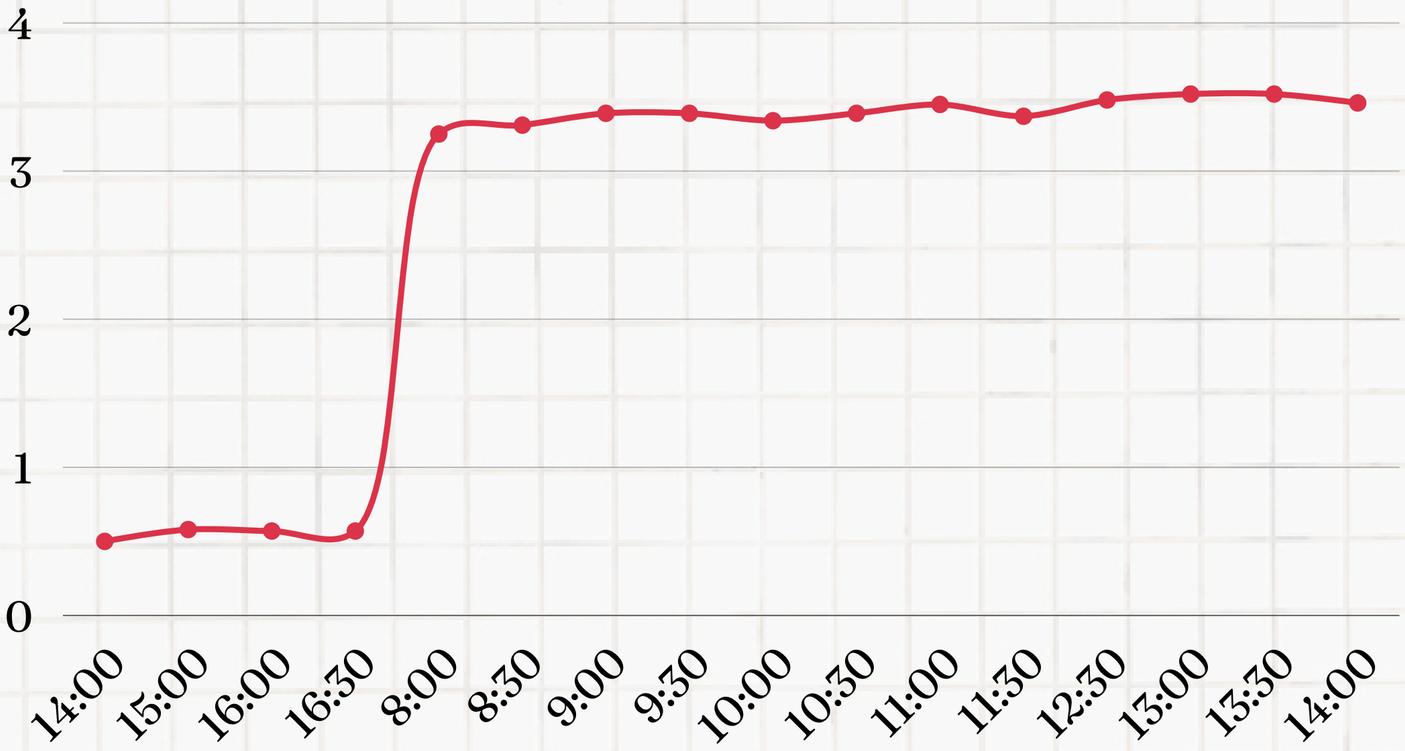


	Strains 1	Strains 4	Strains 5
Hour 0 (14:00)	0,5	0,6	0,5
Hour 1 (15:00)	0,58	0,66	0,48
Hour 2 (16:00)	0,57	0,71	0,49
Hour 2 + 30' (16:30)	0,57	0,77	0,48
Hour 18 (8:00)	3,25	3,58	1,95
Hour 18 + 30' (8:30)	3,31	3,66	2,13
Hour 19 (9:00)	3,39	3,73	2,13
Hour 19 + 30' (19:30)	3,39	3,76	2,33
Hour 20 (10:00)	3,34	3,76	2,32
Hour 20 + 30' (10:30)	3,39	3,83	2,31
Hour 21 (11:00)	3,45	3,81	2,28
Hour 21 + 30' (11:30)	3,37	3,86	2,36
Hour 22 + 30' (12:30)	3,48	4	2,35
Hour 23 (13:00)	3,52	4,05	2,32
Hour 23 + 30' (13:30)	3,52	4,1	2,27
Giờ số 24 (14:00)	3,46	4,06	2,27

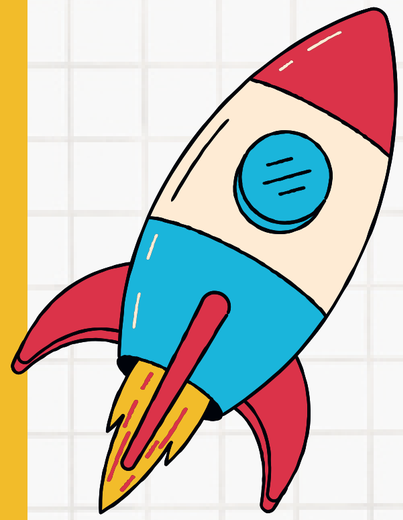
TURBIDITY OVER TIME CHARTS FOR SPECIES 1

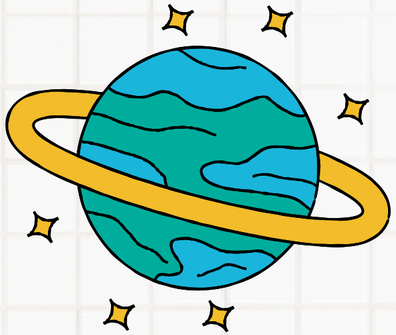


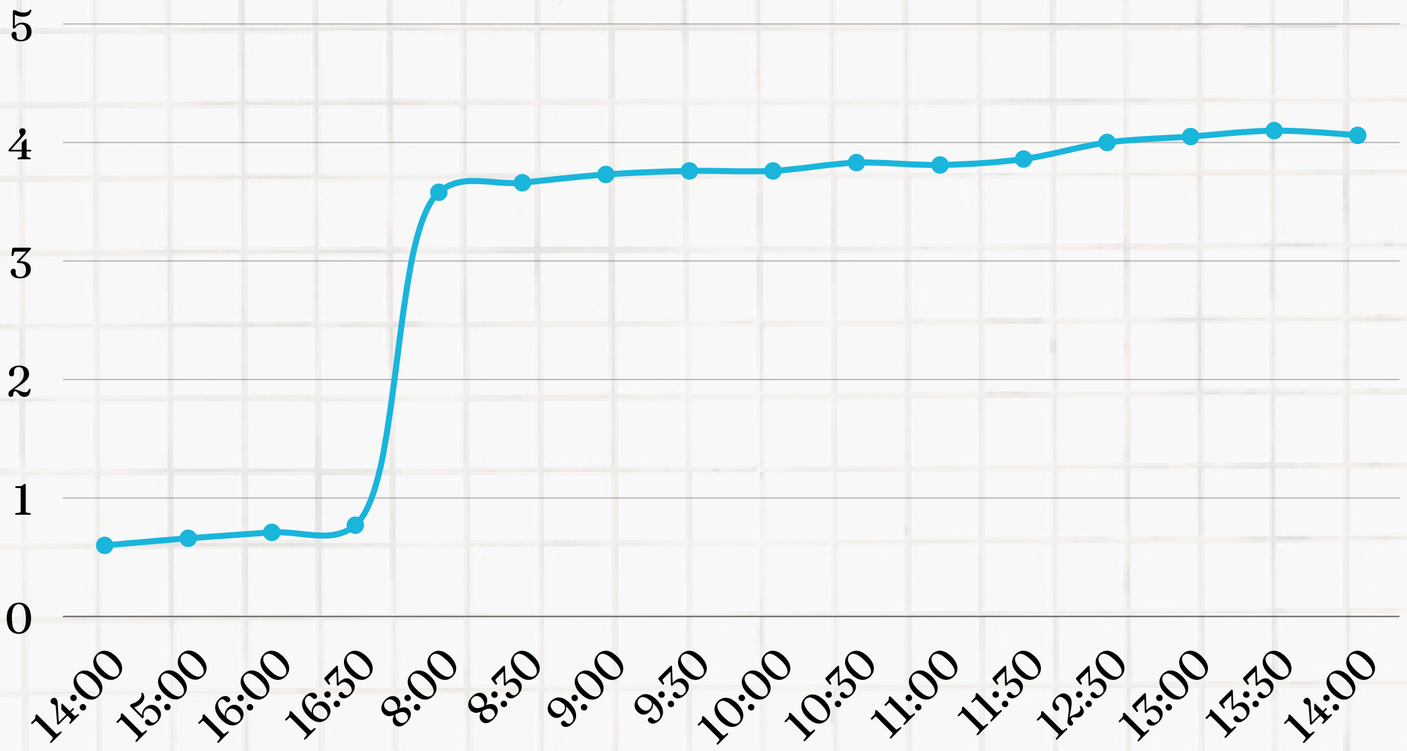




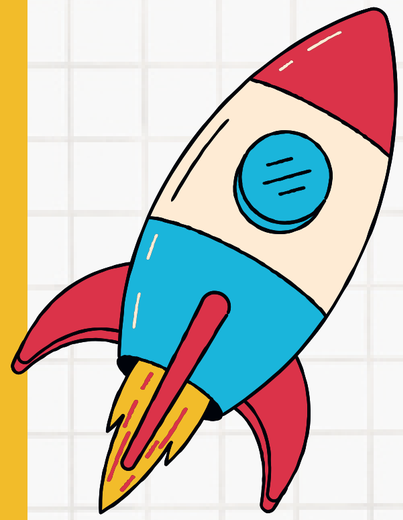
TURBIDITY OVER TIME CHARTS FOR SPECIES 4

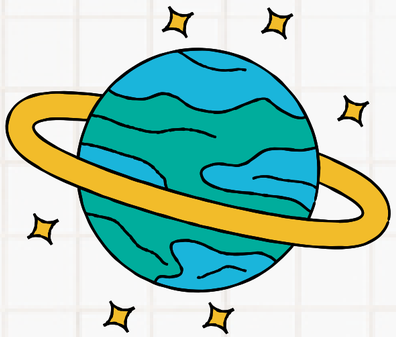


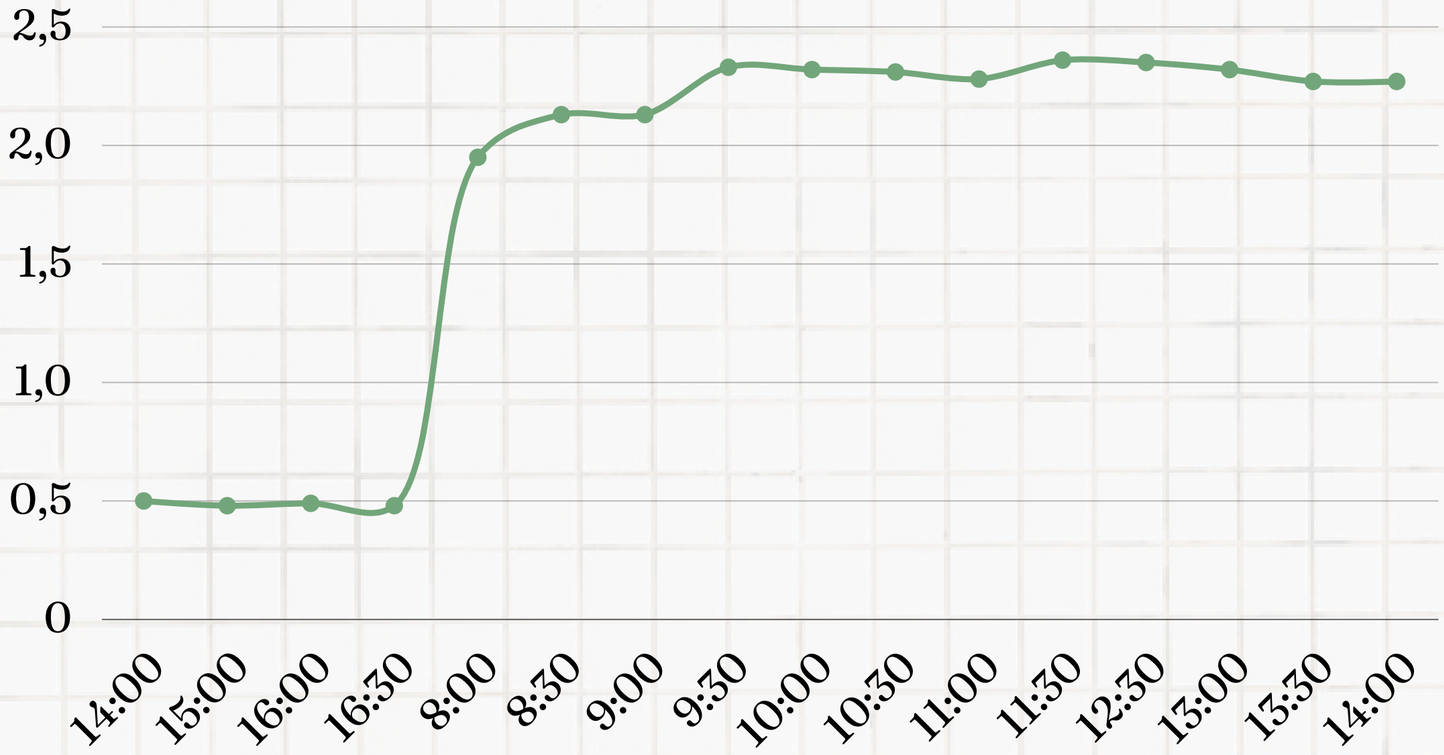




TURBIDITY OVER TIME CHARTS FOR SPECIES 5







THANK YOU!

